

BLACK SEA TECHNOLOGY · COMPLETE PRODUCT REFERENCE

High-Speed Interconnect

Product Specifications

400G / 800G / 1.6T optical modules, high-speed copper cables, active optical cables, and fiber patch cords — ISO 9001:2015 certified, IEEE and CMIS compliant, engineered for AI clusters, hyperscale data centers, and HPC environments worldwide.

1.6T PEAK PORT BANDWIDTH	40+ SKUS ACROSS 6 CATEGORIES	ISO 9001 QUALITY CERTIFIED	2-Year PRODUCT WARRANTY
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OPTICAL INTERCONNECT · NEXT-GENERATION

1.6T Optical Module Series

1.6T OSFP DR8 IHS Transceiver

The flagship next-generation interconnect module. Adopts an OSFP DR8 IHS package with 1600 Gbps single-port bandwidth using PAM4 modulation. Engineered for high-density data centers and AI training clusters. Hot-pluggable with DDMI digital diagnostics. RoHS 2.0 compliant. Fully compatible with mainstream switching equipment.

AI TRAINING & INFERENCE

SUPERCOMPUTING / HPC

HIGH-DENSITY DATA CENTERS

1600G ETHERNET



TECHNICAL PARAMETERS

Parameter	Specification
TRANSMISSION RATE	1600 Gbps · 8 × 106.25 GBd PAM4
FORM FACTOR	OSFP DR8 IHS · Hot-pluggable
POWER CONSUMPTION	≤ 25 W
OPERATING TEMPERATURE	0 °C – 70 °C
TRANSMISSION DISTANCE	500 m on G.652 single-mode fiber
CORE FUNCTIONS	DDMI digital diagnosis · Hot-pluggable
COMPLIANCE	ISO 9001:2015 · IEEE · CMIS · RoHS 2.0

Manufactured by Black Sea Technology Pte. Ltd. Specifications subject to change. Contact sales@innovaas.co for current datasheet.

KEY ADVANTAGES

Advantage	Detail
HIGH BANDWIDTH	1600 Gbps rate eliminates network bottlenecks in the most demanding AI training workloads
LOW POWER	≤ 25 W stable operation in high-temperature, high-density data center environments
COMPATIBILITY	Compliant with international standards; compatible with mainstream equipment; plug-and-play deployment

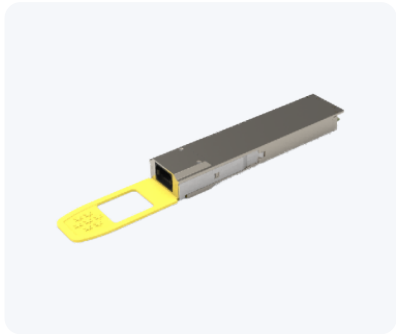
OPTICAL INTERCONNECT · HIGH-DENSITY

800G Optical Module Series

800G OSFP DR4 & DR8 Transceivers

Dual OSFP form-factor 800G modules for AI computing centers, GPU cluster interconnect, and large cloud data centers. Both models adopt PAM4 modulation with hot-plug support and DDMI diagnostics, targeting short-to-medium reach across spine-leaf and rack-to-rack fabric runs.

- AI COMPUTING CENTERS
- SUPERCOMPUTING / HPC
- CLOUD DATA CENTERS
- GPU CLUSTER INTERCONNECT



TECHNICAL PARAMETERS — DR4 VS DR8

Parameter	OSFP DR4	OSFP DR8
TRANSMISSION RATE	850 Gbps	850 Gbps
MODULATION	4 × 106.25 GBd PAM4	8 × 53.125 GBd PAM4
POWER CONSUMPTION	≤ 18 W	≤ 16 W
TRANSMISSION DISTANCE	500 m on G.652 SMF	500 m on G.652 SMF
CORE FUNCTIONS	DDMI · Hot-pluggable	DDMI · Hot-pluggable
COMPLIANCE	RoHS 2.0 · IEEE · CMIS	RoHS 2.0 · IEEE · CMIS

Manufactured by Black Sea Technology Pte. Ltd. Both models are single-mode, single-fiber-pair designs. Contact sales@innovaas.co for compatibility matrix.

800G ADDITIONAL SKUS

Multimode SR & Active Optical

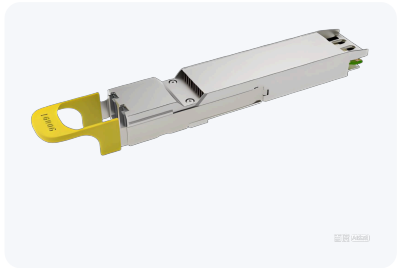
MULTIMODE SR 800G OSFP SR8 Transceiver	ACTIVE OPTICAL CABLE (AOC) 800G OSFP AOC (1 - 50 m)
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OPTICAL INTERCONNECT · 400G

400G Optical Module Series

Full 400G coverage across QSFP112, QSFP-DD, OSFP-RHS, and OSFP form factors. Single-mode DR4 for fabric runs; multimode SR4/SR8 for short-reach rack connections; AOC assemblies for flexible cabling; breakout variants for fan-out to 100G or 200G ports.

- SPINE-LEAF ARCHITECTURE
- AI FABRIC INTERCONNECT
- DCI
- TOP-OF-RACK SWITCHING



Single-Mode Transceivers

- 400G QSFP112 DR4
- 400G OSFP-RHS DR4
- 400G QSFP-DD DR4

Multimode SR Transceivers

- 400G QSFP-DD SR8
- 400G QSFP112 SR4
- 400G OSFP SR4

Active Optical Cables (AOC)

- 400G QSFP-DD AOC
- 400G QSFP112 AOC
- 400G OSFP AOC

Breakout Assemblies

- 400G QSFP-DD → 4 × 100G QSFP28
- 400G QSFP-DD → 2 × 200G QSFP56

SINGLE-MODE REFERENCE — ALL SPEEDS

Speed	SKU	Form Factor
1.6T	1.6T DR8 OSFP Transceiver	OSFP DR8 IHS
800G DR4	800G OSFP DR4 Transceiver	OSFP
800G DR8	800G OSFP DR8 Transceiver	OSFP
400G	400G QSFP112 DR4 · 400G OSFP-RHS DR4 · 400G QSFP-DD DR4	Multi

OPTICAL INTERCONNECT · LEGACY MULTIMODE

25G – 200G Optical Module Series

Field-proven SFP28, QSFP+, QSFP28, and QSFP56 transceivers and AOC assemblies from 25G through 200G. SR variants for short-reach data center connections; AOC for flexible rack-to-rack cabling; breakout assemblies for fan-out to legacy host interfaces. Full compatibility with international standards.

- LEGACY INFRASTRUCTURE UPGRADE
- TOR FAN-OUT
- MIXED-SPEED FABRIC
- CAMPUS & EDGE DCS



SR Transceivers

25G SFP28 SR	40G QSFP+ SR	100G QSFP28 SR4	200G QSFP56 SR4
400G QSFP-DD SR8	400G QSFP112 SR4	400G OSFP SR4	800G OSFP SR8

Active Optical Cables (AOC)

25G SFP28 AOC	40G QSFP+ AOC	100G QSFP28 AOC	200G QSFP56 AOC
400G QSFP-DD AOC	400G QSFP112 AOC	400G OSFP AOC	800G OSFP AOC

Breakout Assemblies

40G QSFP+ → 4 × 10G SFP+	100G QSFP28 → 4 × 25G SFP28
200G QSFP56 → 2 × 100G QSFP28	200G QSFP56 → 8 × 25G SFP28
400G QSFP-DD → 4 × 100G QSFP28	400G QSFP-DD → 2 × 200G QSFP56

Compatibility: All modules comply with international industry standards and ship with complete test reports. Contact sales@innovaas.co for compatibility matrices, qualification reports, or custom BoM generation.

ELECTRICAL INTERCONNECT

High-Speed Copper — DAC / ACC / AEC



TECHNICAL PARAMETERS

Cable Type	Max Rate	Reach	Power / End	Signal Conditioning
PASSIVE DAC	800 Gbps	0.5 – 1.5 m	0.005 W	None
ACTIVE ACC	1.6 T	0.5 – 2.5 m	1.2 W	Linear re-driver
ACTIVE AEC	1.6 T	1 – 5 m	12 W	Full retimer / CDR
ACTIVE AOC	800 Gbps	1 – 50 m	12 W	Optical (multimode)

All copper assemblies use OSFP form factor. Contact sales@innovaas.co for reach-vs-power selection guidance.

COMPLETE SKU CATALOG

DAC — PASSIVE DIRECT ATTACH	ACC — ACTIVE COPPER CABLE	AEC — ACTIVE ELECTRICAL CABLE
100G DAC	800G ACC	400G AEC
100G Breakout DAC	800G Breakout ACC	800G AEC
200G DAC	1.6T ACC	800G Breakout AEC
200G Breakout DAC	1.6T Breakout ACC	1.6T AEC
400G DAC		1.6T Breakout AEC
400G Breakout DAC		
800G DAC		

Selection guide: DAC for lowest cost/power at ≤ 1.5 m · ACC for ≤ 2.5 m with minimal overhead · AEC for ≤ 5 m where reach matters · AOC for 1 – 50 m runs

OPTICAL INTERCONNECT

Fiber Patch Cords & Custom Assemblies

MPO ↔ MPO and MPO ↔ LC patch cords in single-mode (OS2) and multimode (OM3/OM4) variants. All assemblies are fully customizable to wavelength, length, and connector specification.

SINGLE-MODE (OS2)

Single-Mode MPO ↔ MPO

Single-Mode MPO ↔ LC

MULTIMODE (OM3/OM4)

Multimode MPO ↔ MPO

Multimode MPO ↔ LC



QUALITY & CERTIFICATION

ISO 9001:2015

Full lifecycle quality management system

IEEE

International electrical & electronics standards

CMIS

Common Management Interface Specification

RoHS 2.0

EU environmental compliance

SERVICE & SUPPORT

2-Year Product Warranty

Full manufacturer warranty on all products. Non-human damage covered — free repair or advance replacement.

7 × 24 Technical Support

Around-the-clock professional support for consultations and problem troubleshooting.

Solution Architecture Consulting

Fabric topology design, SKU selection, BoM generation, and compatibility validation — at no extra cost.

After-Sales Tracking

Regular deployment tracking with continuous product and service optimisation over the lifecycle.

CONTACT INNOVAAS SOLUTIONS

SALES ENQUIRIES

sales@innovaas.co

WEBSITE

innovaas.co

DATA CENTER CONNECTIVITY

innovaas.co/solutions/data-center-connectivity

SOLUTION ARCHITECTURE

Included with every enquiry

RESPONSE TIME

Within one business day